Name: …… Student ID: …….

The University of New South Wales

School of Electrical Engineering and Telecommunications

Final Examination (Sample Paper)

Term 3, 2025

ELEC3104: Digital Signal Processing

Candidates should attempt the questions as per the following instructions:

1. This exam constitutes 50% of the total course assessments.
2. **Time allowed:** 2 hours (+ 10 minutes of reading time)
3. You are required to answer only **4 questions**. However, if you answer more than 4 questions (which is NOT recommended), we will select the best 4 questions.
4. **All** questions are presented in the Tiered Learning Framework (TLF) form:
   - The exam will be in the following layout:
     - ❖ Four questions under TLF - Level 2 (4 x16 marks)
     - ❖ One question under TLF - Level 3 (26 marks)
     - ❖ One question under TLF - Level 4 (26 marks)
     - ❖ One question under TLF - Level 5 (32 marks)
   - In doing the exam, you can choose which questions you want to complete, depending on what Level you are aiming for:
     - ❖ If you are aiming to achieve **Level 2**, you will need to **answer all four questions under Level 2**. The maximum mark you can achieve is **64.**
     - ❖ If you are aiming to achieve **Level 3**, you will need to answer **any three Level 2 questions and one Level 3 question**. The maximum mark you can achieve is **74**.
     - ❖ If you are aiming to achieve **Level 4**, you will need to answer **any two Level 2 questions, one Level 3 question and one Level 4 question**. The maximum mark you can achieve is **84**.
     - ❖ If you are aiming to achieve **Level 5**, you will need to **answer any one of the Level 2 questions, one Level 3 question, one Level 4 question and one Level 5 question**. The maximum mark you can achieve is **100**.
5. Print your name, student ID and question number on the front page of each **answer** page. Use a different answer booklet for each question you answer.
6. The questions are NOT of equal value. Part marks are as indicated.
7. The examination paper may **NOT** be retained by the candidate.
8. **Authorised examination materials:** ONE A4 sheet with handwritten notes on both sides. It does NOT need to be handed in with the exam material.
9. The use of UNSW approved calculators is permitted.
10. All answers must be written in ink with legible handwriting.
11. This paper contains 6 pages including this instruction page.

TLF – Level 2: Question 1 [16 marks]

(a) A digital communication link carries binary-coded words representing samples of an input signal

$$x(t) = 5\cos(600\pi t) + 4\cos(1800\pi t)$$

The link is operated at 10,000 bits/s and each input sample is quantised into 1024 different voltage levels.

(i) What are the frequencies in the resulting discrete-time signal x[n]? **[3 marks]**

(ii) If x[n] is passed through an ideal D/A converter what is the reconstructed signal y(t)? **[3 marks]**

(b) Consider a signal, $s(t)$, with a magnitude spectrum satisfying the following equation

$$|\mathrm{S(f)}| = \begin{cases} 3 - |\mathrm{f}| & 2\mathrm{kHz} < |\mathrm{f}| < 3\mathrm{kHz} \\ 0 & \text{otherwise} \end{cases}$$

f – frequency in kHz. The signal *s(t)* is sampled uniformly with a sampling frequency of 2 kHz. Sketch the digital magnitude spectrum. **[6 marks]**

(c) A signal of $v(t)$= 4.5 sin(1000πt) Volts is sampled at 2.5 kHz. An eight-bit A/D converter with a nominal input range of ±5 Volts is used to convert sampled voltages to integer codes.

(i) Calculate the quantization interval of the A/D converter. **[2 marks]**

(ii) Determine the maximum quantization error. **[2 marks]**

TLF– Level 2: Question 2 [16 marks]

(a) A two-pole lowpass filter has the transfer function

$$H(z) = \frac{k_0}{(1 - b_1 z^{-1})^2}$$

(i) Determine k_0, so that $|H(\theta)|$ is equal to 1, when $\theta = 0$. θ is the digital frequency. **[2 Marks]**

(ii) Determine the value of b_1 such that the frequency response $H(\theta)$ satisfies the condition $|H(\theta)|^2_{\theta=\frac{\pi}{4}} = \frac{1}{2}$ **[6 Marks]**

(iii) Find the impulse response of the digital filter $H(z) = \frac{\alpha+\alpha z^{-1}}{1-\alpha z^{-1}}$ **[4 marks]**

(b) Three systems with impulse responses $h_1[n] = \delta[n] - \delta[n-1]$, $h_2[n] = h_1[n]$ and $h_3[n] = u[n]$ are connected in cascade. Calculate the impulse response, $h_c[n]$, of the overall system? **[4 Marks]**

TLF– Level 2: Question 3 [16 marks]

(a) (i) Design a notch filter $H(z)$ with notch frequency 60Hz, stopband 5 Hz (bandwidth) and sampling frequency 300Hz. **[6 marks]**

(ii) Draw an implementation of $H(z)$. **[3 marks]**

(b) Determine the transfer function H(z) of a discrete - time system with the pole-zero map shown in Figure 1. **[3 Marks]**

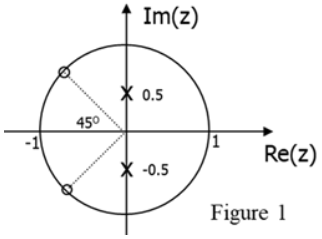


Figure 1

(c) Write the equation for the 4-point DFT, H[k], of the impulse response of a filter h[n] = [1, 2, 3] and what is the DC gain of the filter. **[4 Marks]**

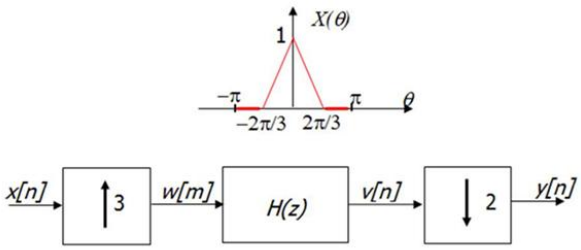
TLF – Level 2: Question 4 (16 marks)

(a) A signal (with a bandwidth of 0 to 3kHz) has been sampled at 8 kHz and for a particular application this signal is required to be used at an increased sampling rate of 18 kHz:

(i) Draw a block diagram to show how you would efficiently implement a system to achieve this, indicating the type of digital filter you would use (give reasons); **[3 Marks]**

(ii) Calculate the minimum cut-off frequency of the digital lowpass filter in terms of rad/sec. **[2 Marks]**

(b) A signal x[n] has a spectrum X(θ) as shown below. The signal x[n] is applied to the system shown below. The ideal lowpass filter H(z) has a gain factor of 1 in the passband and a cut-off frequency θ = π/3. Sketch |X(θ)|, |W(θ)|, |V(θ)|, |Y(θ)| against θ. **[6 Marks]**



(c) Using the following difference equation, you are required to implement (i.e.draw a structure) a digital oscillator that generates y[n] = sin(θ_0n) for n ≥0. Choose appropriate initial conditions. Note that: $x[n]$ is the input; $y[n]$ is the output and r is a constant. You are required to choose an appropriate value for r in order to sustain oscillation

$$y[n] = 2r\cos\theta_0\, y[n-1] - r^2 y[n-2] + x[n] - r\cos\theta_0\, x[n-1]$$

[5 Marks]

End of TLF- Level 2 Examination Questions

TLF – Level 3: Question [26 marks]

(a) The frequency response $H_d(\theta)$ of a linear phase FIR filter is given by:

$$H_d(\theta) = \begin{cases} e^{-j3\theta} & 0 \leq |\theta| \leq \pi/2 \\ 0 & elsewhere \end{cases}$$

(i) Show that the impulse response $h_d[n]$ is given by

$$h_d[n] = \frac{sin\left(\frac{\pi}{2}(n-3)\right)}{\pi(n-3)}$$

[6 marks]

(ii) The length of the impulse response of this linear phase FIR filter is set to 7. Find the values of the FIR filter coefficients using a rectangular window function.

[4 marks]

(iii) Sketch the phase response of the system $H_d(\theta)$.

[4 marks]

(b) (i) Use the bilinear transform to design an IIR digital lowpass filter based on a prototype normalised function:

$$G(s) = \frac{1}{(s^2+s+1)}$$

The digital filter is to be designed to operate at a sampling rate of 8 kHz with a 3 dB cut-off frequency at 2 kHz. Show that G(z) reduces to: **[8 marks]**

$$G(z) = \frac{(z+1)^2}{3(z^2+\frac{1}{3})}$$

(ii) Draw an implementation of this filter G(z) as a cascade structure. **[4 marks]**

End of TLF - Level 3 Examination Question.

TLF – Level 4: Question 1 [26 marks]

(a) Using the z-transform table,

(i) Determine the difference equations for the digital oscillators that have impulse responses $h[n]= A\ sin[n\theta_0]\ u[n]$ and $h[n] = A\ cos[n\theta_0]\ u[n]$ respectively.

[8 Marks]

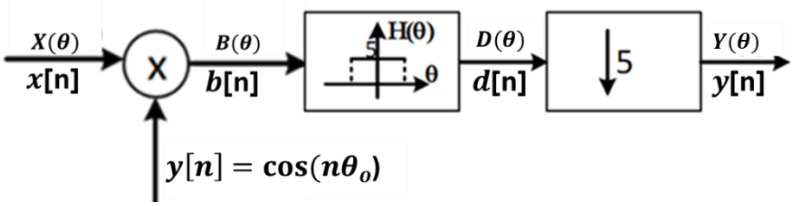
(ii) Determine the structure for the coupled-form oscillator by combining the difference equations obtained in part (i) above.

[6 Marks]

(b) A The input $X(\theta)$ for the digital modulator shown below is given by:

$$X(\theta) = \begin{cases} 1 & 0.4\pi \le |\theta| \le 0.8\pi \\ 0 & elsewhere \end{cases}$$

The modulator signal frequency is $\theta_o = 0.6\pi$. $H(\theta)$ is an ideal lowpass filter with a cut-off frequency of 0.2π and a gain of 5. Sketch the magnitude spectrum of $x[n], b[n], d[n]$ and $y[n]$, one above the other. **[12 Marks]**

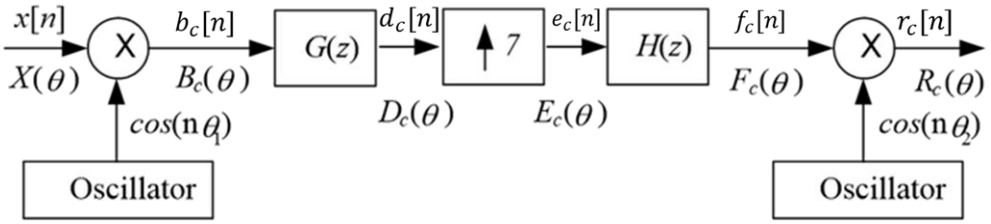


End of TLF- Level 4 Examination Question.

TLF – Level 5: Question [32 marks]

(a) The diagram below shows a digital modulator. The sampling period T of the input signal x[n] is 125 μs. The first oscillator generates a carrier of frequency θ_1. The second oscillator generates a carrier of frequency θ_2: $\theta_1 = \omega_1 \frac{T}{2}$ and $\theta_2 = (\omega_c + \omega_1)\frac{T}{2}$;
where $\omega_1 = 2\pi \,.\, 2 \,.\, 10^3\ rad/sec$ and $\omega_c = 2\pi \,.\, 4 \,.\, 10^3\ rad/sec$
The low-pass filters G(θ) and H(θ) have the following characteristics:

$$G(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \frac{\pi}{4} \\ 0 & \frac{\pi}{4} < |\theta| \le \pi \end{cases} \qquad H(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \frac{\pi}{28} \\ 0 & \frac{\pi}{28} < |\theta| \le \pi \end{cases}$$



(i) If $X(\theta)$ = 1+cos(2θ), by writing appropriate equations for $B_c(\theta), D_c(\theta), E_c(\theta), F_c(\theta)$, show that

$$R_c(\theta) = \begin{cases} \frac{1}{2} & \frac{10\pi}{14} \le \theta \le \frac{11\pi}{14} \\ \frac{1}{2} & \frac{-10\pi}{14} \le \theta \le \frac{-11\pi}{14} \\ 0 & elsewhere \end{cases}$$

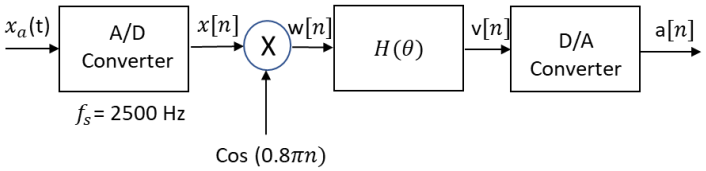
[8 marks]

(ii) Sketch one above another, $|Dc(\theta)|$, $|Ec(\theta)|$, $|Fc(\theta)|$and $|Rc(\theta)|$ against θ in the fundamental frequency region $-\pi \le \theta \le \pi$ **[8 marks]**

(b) An analogue signal of the form x_a(t) = a(t) cos2000πt is bandlimited to the range 900≤f≤1100 Hz. It is used as an input to the system shown below:

(i) Determine and sketch the spectra for the signal x[n] & w[n]. **[8 marks]**

(ii) Use a rectangular window of length M = 7 to design a lowpass linear phase FIR filter $H(\theta)$ that passes a[n]. **[8 marks]**



End of TLF - Level 5 Examination Question.

End of Examination